# HTDFormer: Hyperspectral Target Detection Based on Transformer With Distributed Learning

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***Abstract*— In recent years, many hyperspectral target detection (HTD) methods based on advanced techniques have been proposed and achieved good results. However, the large amount of data produced by satellites and airborne remote sensing instruments has posed new challenges for efficient target detection of massive hyperspectral images (HSIs). In this article, we propose a new weakly supervised HTD framework based on a transformer with distributed learning (HTDFormer), which capitalizes on the parallel processing capabilities of multiple workers to efficiently handle large-scale HSIs. Specifically, the HTDFormer framework effectively integrates both spectral and spatial features within a unified optimization procedure via the transformer mechanism. A flexible sample augmentation approach is proposed to overcome the limitations of inadequate well-labeled training instances and meet the requirements of the transformer. To facilitate model training, we introduce the concept of distributed deep learning (DDL) into HTDFormer by leveraging a ring all-reduce (RAR) decentralized architecture, which embeds distributed learning into an HTD framework for the first time. Furthermore, the large-batch training strategy and the gradient compression strategy are used to enable large-scale distributed processing and reduce communication costs, respectively. Finally, an exponentially constrained nonlinear function is adopted to acquire pixel-level prediction via spectral–spatial fusion. Experimental results demonstrate that the proposed framework achieves promising performance with regard to the increasing scale of real HSIs.**

***Index Terms*— Distributed learning, hyperspectral target detection (HTD), sample augmentation, transformer.**

## I. Introduction

HYPERSPECTRAL imagery sensors usually simultaneously acquire images of the same scene in many contiguous and narrow spectral bands [1]. Due to the high spectral resolution, hyperspectral images (HSIs) with the material discriminative ability are suitable for many applications [2], including classification, target detection, and anomaly detection.

Manuscript received 25 July 2023; revised 19 August 2023 and 12 September 2023; accepted 15 September 2023. Date of publication 19 September 2023; date of current version 29 September 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62121001, Grant 62371365, Grant 62322117, and Grant U22B2014; in part by the Youth Talent Promotion Project of China Association for Science and Technology under Grant 2020QNRC001; and in part by the Fundamental Research Funds for the Central Universities and the Innovation Fund of Xidian University under Grant YJSJ23012. *(Corresponding author: Weiying Xie.)*

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Digital Object Identifier 10.1109/TGRS.2023.3317033

By exploiting spectral features, hyperspectral target detection (HTD) is aimed at automatically extracting targets of interest from remote sensing images, which is significant for both military and civilian applications. The HTD technique can be regarded as a binary classifier that extracts small targets from large unknown background. In HSIs, whether two materials belong to the same category depends on their spectral characteristic which is difficult with the presence of interference and spectral aliasing [3].

A great number of HTD methods have been proposed in the past decades, which can be grouped into two major categories: traditional and learning-based HTD.

As for traditional models, deterministic detection algorithms based on spectral similarity, such as spectral angle mapping (SAM) [4], measure the distance between targets and background pixels by calculating their spectral difference. Structured background approaches assume that any mixed pixel can be regarded as a linear combination of several endmembers with different abundances [5]. The orthogonal subspace projection (OSP) [6] model is representative. Detectors based on the unstructured background model are derived from a generalized likelihood ratio test, such as the adaptive cosine/coherent estimator (ACE) [7] which models background with a generic distribution. Some methods are based on a statistical index such as constrained energy minimization (CEM) [8] and matched filter (MF) [9] in which the second-order statistic is applied. Besides, many variants and extended versions based on conventional models have been developed for better performance. For example, as the most representative statistic-based HTD algorithm, CEM has been widely researched and its variants were proposed such as hierarchical CEM (hCEM) [10] and ensemble-based cascaded CEM (E-CEM) [11].

However, there are still shortcomings in the existing models. First, these algorithms define the target spectral characteristics via either a single target spectrum or a target subspace ideally. The former models the background statistically by a Gaussian distribution, while the latter relies on a subspace to represent the whole or the local background statistics [12]. Second, some conventional approaches are heavily dependent on handcrafted features. Most of these shallow features rely on expert knowledge limited by human factors. Hence, these methods cannot effectively extract discriminative information for real HSIs which change considerably with the imaging environment [13]. Third, some traditional methods usually require rigorous assumptions on the spectral distribution, which is not

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fully realistic for real HSIs. For instance, the ACE assumes that the background has additional noise, and the OSP assumes that the noise is a zero-mean multivariate normal distribution.

Recently, given the superior data analysis capabilities of machine learning (ML) and deep learning (DL), learning-based methods are increasingly used in the field of remote sensing image analysis. The learning-based methods can capture nonlinear and deep features and achieve complex function approximation through layer-by-layer combination [14], which has achieved success in computer vision and has attracted the attention of HSI processing.

Kernel-based ML as well as sparse and collaborative representation are the most typical ML approaches [15]. The kernel-based methods further extend the approaches above-mentioned to nonlinear versions and solve nonlinear problems. For example, a kernel OSP maps the original space into the kernel space for handling the problem of linear inseparability [16]. However, how to design a suitable kernel remains a problem. As for sparse and collaborative representations, many methods have been developed, such as a sparsity-based target detector (STD) [17] and combined sparse and collaborative representation (CSCR) [18]. To meet the requirements of dictionary construction, the representation-based methods need more information about target signatures. If the prior target information is insufficient, the performance of such detectors would be severely degraded [15].

Under a data-driven manner, DL-based methods can capture nonlinear and deep features, as well as realize complex function approximation, which has been widely concerned in different applications. On this basis, several DL-based HTD algorithms have been presented to capture nonlinear properties of HSIs via autoencoder (AE) [19] and convolutional neural network (CNN) [20], as well as their variants. There are there main categories of DL-based HTD methods.

1) Methods in a supervised manner: These methods mainly enlarge target samples via synthesis and then construct an end-to-end detector by training on large amounts of pixel pairs [15]. The typical detectors include CNN-based detection (CNND) [21], CNN target detector (CNNTD) [22], HTD-Net [23], two-stream convolutional network-based target detector (TSCNTD) [24], etc.
2) Methods in an unsupervised manner: These methods usually enhance the discrimination capacity with unsupervised networks under certain constraints, and then perform target detection with the simple matching strategy [15]. The representative detectors include spectral regularized unsupervised networks (SRUNs) [25], 3-D macro–micro residual AE detector (3D-MMRAED) [26], deep spatial–spectral network (DSSN) [27], etc.
3) Methods in a weakly supervised manner: These methods have to access label information from unlabeled data to leverage incomplete supervision and alleviate label noise. Under weak supervision, only a small subset of training data is observed with labels while others remain unlabeled. Meanwhile, the given labels may be inaccurate [3]. On this basis, the generative adversarial network (GAN) [28] and meta-learning [29] have been introduced in HTD tasks, to estimate the underlying noise-free distribution with insufficient labeled data.

Recently, some of the advanced DL mechanisms, such as the transformer [30], [31] with spatial–spectral attention [32], [33] or cross attention [34], have been verified for computer vision and introduced in hyperspectral tasks including classification [32], [33], [34], image fusion [35], and video tracking [36], [37]. However, whether they can be used for HTD tasks and perform well has not been fully investigated.

The above learning-based HTD methods generally obtain better detection performance than traditional methods. However, there are three aspects of algorithms that need to be optimized as follows. The first limitation is the vulnerability of supervised detection to training samples. It is challenging to acquire plentiful labeled training samples in practice. Moreover, these methods may have a certain effect on training sample enlargement, but there is still a shortage of using global spatial information. Models using only spectral information are not sufficient to effectively distinguish ground targets. Besides, models based on CNN or the fully convolutional network (FCN) [38] are severely constrained by expensive computation complexity and sensor-specific transferability.

In addition to the above problems, the tremendous amount of remotely sensed data has posed serious challenges to the efficient and scalable processing of remotely sensed data in the context of various applications [39]. As hyperspectral data volume continues to increase, computational efficiency is of greater importance to processing massive HSIs by using the above advanced techniques.

In HSIs, each pixel of the collected data cube can be represented by a vector of which the entries correspond to the spectral bands, providing detailed spectral information of the underlying materials within the pixel. Due to the large number of bands contained in HSIs, the processing of high-dimensional HSI data generally requires huge storage space and heavy computational load, and therefore falls within the category of big data problems [40].

Despite the promising discriminative capability, DL-based HTD methods in general have very high computational complexity during training especially when computing attention maps and would suffer from extremely long computation time when processing large-scale hyperspectral datasets.

On this basis, to cope with the big data challenge in HTD tasks and make full use of global spatial information in HSIs, we intend to design a united HTD framework from the following two perspectives.

1) In reality, the spatial contexts of HSIs can provide complementary information to their abundant spectral features for more precise recognition [41]. To make full use of the correlation between the spectral and spatial domains, we intend to introduce the transformer [30], [31] mechanism into HTD tasks. As an advanced DL mechanism in the natural language processing (NLP) field, the transformer is flexible and easily generalized due to the need of processing a vast variety of sentences produced by the combinatorial generalization of words and grammar. Different from CNN, usually used to model long-distance dependency of image data

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via large receptive fields, the transformer can process and analyze sequential data effectively by acquiring global contextual knowledge through the self-attention mechanism [30], [31].

2) Considering the heavy computational load from DL-based hyperspectral applications, it is necessary to develop solutions to accelerate the processing of large-scale hyperspectral datasets. Fueled by advances in distributed DL (DDL), recent years have witnessed a rapidly growing demand for resource-intensive distributed computing to process DDL computing jobs [42]. Since the acceleration of training through exploiting model parallelism is limited, current research principally focuses on data parallelism [43], wherein each training iteration, each worker has a full copy of model parameters, trains a subset of the input data, exchanges parameter updates (gradients) with each other, and then updates model parameters [44]. Meanwhile, as workers exchange gradients frequently, communication often occupies a significant portion of overall training time [44], which needs to be optimized.

To realize the above ideas, we introduce the transformer [30] mechanism into HTD tasks to extract spectral–spatial features for target–background separation. Considering data parallelism with synchronous parameter update is a popular method for distributed model training with better convergence guarantees [44], we develop a new DDL form that incorporates the ring all-reduce (RAR) decentralized architecture and model training strategy in this work. On this basis, we are able to conduct large-batch training for the proposed transformer-based HTD framework with distributed learning. In the experiment, three hyperspectral datasets with high spectral resolution and high spatial resolution have been used. The contributions of this article can be summarized as follows.

1) A novel weakly supervised HTD framework based on the transformer, namely, HTDFormer, has been proposed in this work. HTDFormer combines spectral and spatial features into a unified optimization procedure, which solves the problem of insufficient utilization of global spatial information. To address the issue of lacking well-labeled training instances and meet the requirement of the transformer mechanism, the sample augmentation stage is designed in a flexible manner.
2) To enable HTD processes for massive hyperspectral data, we describe a new DDL form in HTDFormer, which speeds up transformer training via a data-parallel strategy. By means of ring-all-reduce decentralized architecture, HTDFormer takes advantage of the parallel processing abilities of workers to cope with large-scale HSIs. As far as we know, this is the first time that the DDL mechanism has been embedded into an HTD framework, which opens up more flexible ways for HTD tasks.
3) To further accelerate model training, the large-batch training strategy has been adopted to realize large-scale distributed learning. Increasing the batch size enables the framework to scale to more workers without reducing the workload on each worker. Considering large-scale distributed training requires significant communication bandwidth for gradient exchange, we use the gradient compression strategy to reduce communication cost without changing the model structure.

The remainder of the article is organized as follows. Section II introduces the detailed framework of HTDFormer. Section III presents the experimental results and corresponding analysis. Finally, the conclusion is drawn in Section IV.

## II. HTDFormer Framework

The proposed HTDFormer consists of three stages: sample augmentation, distributed training, and spectral–spatial testing. Sample augmentation involves converting primitive hyperspectral pixels into spectral sequences for weakly supervised learning. This stage includes coarse detection and sequence construction based on classic detectors. The training stage uses the spectral sequences to train the transformer encoder, which captures the discriminative spectral–spatial similarity between hyperspectral pixels and their neighboring pixels. Distributed learning with large-batch training and gradient compression accelerates transformer training, especially for large-scale HSIs. In the testing stage, joint spectral–spatial inference is applied to combine the discriminative features from the spectral and spatial domains.

### A. Sample Augmentation

In HTD tasks, current spectroscopic detectors often result in a low target presence in large-scale hyperspectral scenes. The prior knowledge for the HSIs usually consists of a single target spectrum, known as the target signature. To provide sufficient training samples, a sample augmentation procedure is proposed. In this procedure, coarse detection is realized via an efficient classic detector, such as ACE [7] or CEM [8], which provides essential information for subsequent spatial–spectral feature extraction. The coarse detector used in this work can also be substituted with other widely used detectors.

*1) Coarse Detection:* Considering ACE has shown great success in HTD tasks [7], [45], [46], we adopted ACE as a coarse detector to construct pseudo datasets with adequate samples for training. As a spectral match-based detector, ACE makes decisions based on signatures to distinguish the target pixels from the background by comparing the whole image spectra with the known target spectra [46], i.e., the signature.

Let an $N$-pixel HSI be denoted as $\mathbf{X} \in \mathbb{R}^{H\times W\times L}$ consisting of spectrums with L-bands. Each spectrum can be denoted as a vector $\mathbf{x}(i) = [x_1(i), \ldots, x_L(i)]^T (i = 1, \ldots, N)$. All the spectrums can be arranged in the matrix form as $\mathbf{X} = [\mathbf{x}(1), \ldots, \mathbf{x}(N)]$. Let $\mathbf{d} = [d_1, \ldots, d_L]^T$ represent the spectral signature of the target of interest, which can be obtained from the spectral library [45] and basically represent the average distribution of real target samples of a specific category.

For a pixel $\mathbf{x}$, the output of the ACE is

$$y_{\mathrm{ACE}}(\mathbf{x}) = \cos^2\theta = \frac{\left(\tilde{\mathbf{x}}^T\tilde{\mathbf{d}}\right)^2}{\|\tilde{\mathbf{x}}\|^2\|\tilde{\mathbf{d}}\|^2} \tag{1}$$

where $\theta$ is the spectral angle between $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{d}}$. And $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{d}}$ can be obtained via the whitening transform as [46]

$$\tilde{\mathbf{x}} = \hat{\mathbf{\Gamma}}^{-\frac{1}{2}}\mathbf{x}_0 \quad \tilde{\mathbf{d}} = \hat{\mathbf{\Gamma}}^{-\frac{1}{2}}\mathbf{d}_0 \tag{2}$$

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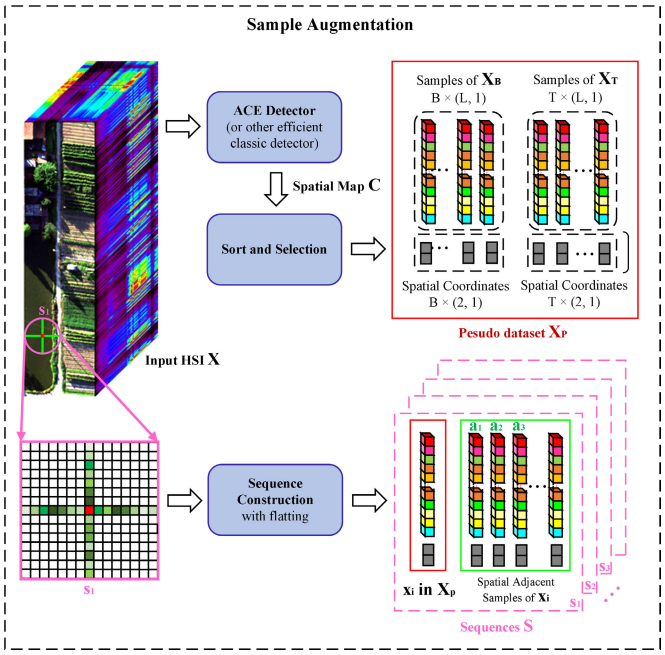



Fig. 1. Sample augmentation stage produces the training set $\mathbf{S}$, in which each sequence $\mathbf{s}_i$ consists of the central spectrum $\mathbf{x}_i$ from pseudo dataset $\mathbf{X}_P$ and $j$ spectrums $\mathbf{a}_{ij}$ surrounding $\mathbf{x}_i$.

where $\mathbf{x}_0$ and $\mathbf{d}_0$ represent the mean-removed spectrum $\mathbf{x}$ and signature $\mathbf{d}$ [46], and $\hat{\mathbf{\Gamma}} = (1/N)\sum_{i=1}^{N}[\mathbf{x}_0(i)\mathbf{x}_0^T(i)]$ is the estimated covariance matrix.

Therefore, the result $\mathbf{C}$ of coarse detection using the ACE detector can be regarded as a spatial map, where each pixel represents the likelihood of a certain spectrum being a target. Then we sort the pixels of $\mathbf{X}$ based on the value of each corresponding pixel in spatial map $\mathbf{C}$, as shown in Fig. 1. With reference to the distribution of real target pixels and real background pixels in the real HSIs, we select the last $\sigma\%$ samples to form the pseudo target dataset $\mathbf{X}_T \in \mathbb{R}^{T\times L}$, and select the top 5% samples with relatively small values to form the pseudo background dataset $\mathbf{X}_B \in \mathbb{R}^{B\times L}$. Thus, we can obtain the pseudo dataset $\mathbf{X}_P$ consisting of spectrums in $\mathbf{X}_B$ and $\mathbf{X}_T$, for subsequent procedures. The intraclass differences in samples in $\mathbf{X}_B$ are large, and the intraclass differences in samples in $\mathbf{X}_T$ are small. Note that the ratio for selecting pseudo background pixels or pseudo target pixels is not fixed. A range of the most appropriate ratio exists for each specific dataset. Besides, it is worth noting that training instances from the pseudo datasets constructed based on coarse detection from limited prior target spectral information may be incompletely and inaccurately labeled. Therefore, the proposed framework can be considered as a method under inaccurate supervision, one of the three types of weak supervision [47].

*2) Sequence Construction:* To ensure that training and testing samples meet the input requirement of transformer mechanism, the sequence construction procedure is used. Specifically, HTDFormer constructs the sequence samples based on spectrums (in $\mathbf{X}_P$) and its adjacent spectrums, as shown in Fig. 1. To make the model efficient and effective, we use a cross region containing a sample in $\mathbf{X}_P$ to produce a sequence sample as an input for transformer training.

Specifically, given a pixel $\mathbf{x}_i$ in $\mathbf{X}_P$ whose class label is to be predicted, a cross region containing $\mathbf{x}_i$ is selected and flattened into a pixel sequence $\mathbf{s}_i$. The cross region is composed of a horizontal line and a vertical line, and both contain 15 pixels. A crossing region contains 30 pixels in total, in which the duplicated pixel $\mathbf{x}$ is not removed for simplification in processing. Therefore, the set of sequence samples $\mathbf{S}$ for transformer training can be obtained based on $\mathbf{X}_P$ and $\mathbf{X}$. Then, each pixel $\mathbf{x}_i$ in $\mathbf{X}_P$ is included in the corresponding pixel sequence $\mathbf{s}_i$ through the multihead self-attention (MHSA) mechanism of transformer [30], [31].

Subsequently, the pixel sequences will be regarded as normal training samples and then fed into the proposed framework based on the transformer for weakly supervised training.

### B. Distributed Training

On one hand, to take full advantage of the global spectral–spatial features of HSIs, the transformer with a global receptive field [30] is applied to HTD tasks in this work. On the other hand, to cope with large sizes of real hyperspectral datasets, it is necessary to realize parallelism during training, in which the model under training is replicated in several compute nodes, i.e., workers, typically equipped with powerful graphics processing units (GPUs) [48]. In HTDFormer, we implement data-parallel distributed training where each worker possesses a local copy of the entire model and independently processes a partition of the whole training samples in a training epoch.

*1) Target Detection With Transformer:* As demonstrated in Fig. 2(a), the proposed HTDFormer leverages a transformer encoder to capture discriminative spectral–spatial similarity of each pixel and its neighboring pixels regardless of their spatial distance. For each pixel $\mathbf{x}$ in an input spectral sequence $\mathbf{s}_i$, the transformer encoder computes an attention distribution over the crossing region, and the attention distribution informs $\mathbf{x}$ where it should focus on.

First, the pixel sequence $\mathbf{s}_i$ is transformed using the spectrum embedding (SE) operation, which transforms pixel vectors to a new vector space with a predefined dimension set as 50. Then, the position embedding (PE) operation is added to the SEs to retain positional information. SE and PE can be formulated as

$$\mathbf{E} = [\mathbf{e}_1; \mathbf{e}_2; \ldots; \mathbf{e}_M] = [\mathbf{s}_1\mathbf{E}_S; \mathbf{s}_2\mathbf{E}_S; \ldots; \mathbf{s}_M\mathbf{E}_S] + \mathbf{E}_P \quad (3)$$

where $\mathbf{E}_S$ represents the learned weights of the linear transformation, and $\mathbf{E}_P$ is the learned positional matrix.

Therefore, for the spectral sequences $\mathbf{S}$, the SE and PE operations allow the transformer encoder with the desired architecture to be used in a flexible input hyperspectral dimension. Besides, the two operations can increase the training speed by reducing the input dimension of HSIs. The resulting sequences $\mathbf{E}$ consisting of embedding vectors $\mathbf{e}_i$ serve as input to the transformer encoder.

The transformer encoder uses MHSA [31] based on the self-attention mechanism whose architecture is the same

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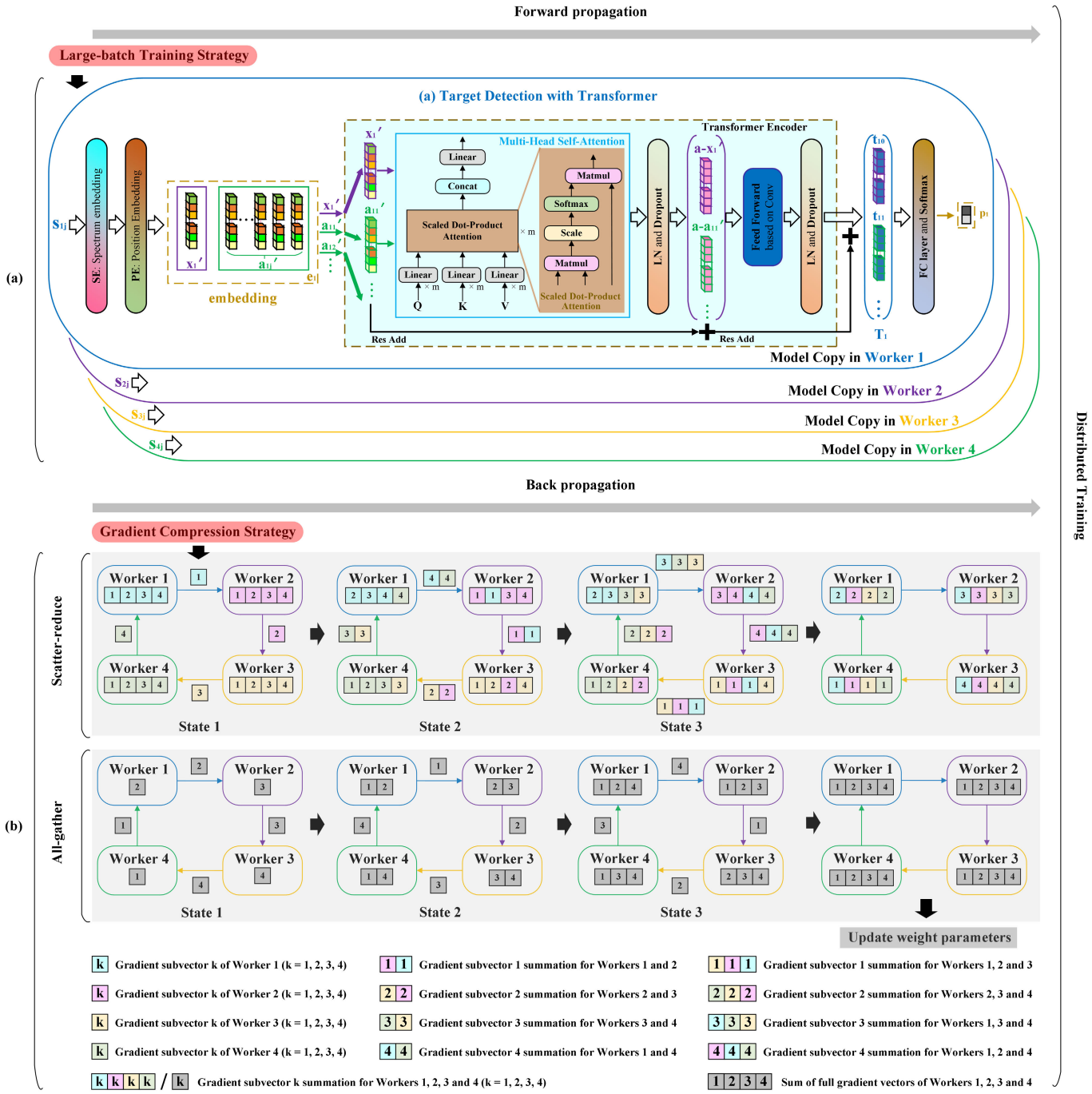



Fig. 2. Distributed training stage consists of two procedures. (a) Target detection with transformer. (b) Distributed learning strategy. **For a single worker**, as shown in (a), each sequence $\mathbf{s}_i$ has been transformed via SE and PE to sequence $\mathbf{e}_i$, which consists of embedded vectors $\mathbf{x}_i'$ and $\mathbf{a}_{ij}'$ with predefined dimension. Then, the MHSA mechanism is conducted in each embedded sequence $\mathbf{e}_i$ to produce learned attention distribution over this sequence. Therefore, each embedded vector in $\mathbf{e}_i$ acquires different attentions via MHSA and was transformed to attention vectors $\mathbf{a}$-$\mathbf{x}_i'$ and $\mathbf{a}$-$\mathbf{a}_{ij}'$ after LN and dropout operation. And the encoded pixel sequence $\mathbf{T}_i$ consisting of $\mathbf{t}_{ij}$ can be obtained through the feedforward module of the transformer encoder based on $\mathbf{e}_i$. On this basis, the output label $\mathbf{p}_i$, the possibility of $\mathbf{x}_i$ to be target or background, can be produced based on $\mathbf{T}_i$ through a fully connected (FC) layer. **For multiple workers**, as shown in (b), the training samples are divided into $n$ partitions where $n$ is the number of workers in the ring and set as four in this article. Each partition will be designated to train a full model copy in a worker. Then, the distributed learning strategy consisting of two steps has been implemented. In the scatter–reduce step, the workers partition the parameters into $n$ parts. Next, the workers will do $n-1$ iterations of the scatter–reduce. In every iteration, each worker sends $1/n$ of its gradients, i.e., a chunk, to its right neighbor and will receive a chunk from its left neighbor. After the first send and receive is completed, each worker will have a chunk that consists of the sum of that same chunk on two different workers. In the next iterations, the process continues, and by the end, each worker will have one chunk that contains the sum of all the values in that chunk across all the workers. In the all-gather step, after the scatter–reduce step is complete, every worker has a final chunk which includes contributions from all the workers. To complete the all-reduce, the workers exchange those chunks for $n-1$ iterations of the all-gather, so that all the workers have all the final chunks.

in [30]. The mechanism can be described as mapping a query and a set of key–value pairs to an output, where the query, keys, values, and output are all vectors. The output is computed as a weighted sum of the values, where the weights

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assigned to each value will be computed by a compatibility function of the query with the corresponding key [30]

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \tag{4}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ represent the query matrix, key matrix, and value matrix, respectively, and $d_k$ is the dimension of the input data. The MHSA with $m$ heads can be formulated as

$$\text{MHSA}(\mathbf{E}) = \text{Concat}(\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_m)\mathbf{W}^O \tag{5}$$

where $\mathbf{h}_i = \text{Attention}(\mathbf{E}\mathbf{W}_i^Q, \mathbf{E}\mathbf{W}_i^K, \mathbf{E}\mathbf{W}_i^V)$, and $\mathbf{W}$ represents the corresponding learned matrix.

The layernorm (LN) [31] operation is applied after MHSA

$$\mathbf{e}_i^{\text{Trm}} = \text{LN}(\text{MHSA}(\mathbf{e}_i)). \tag{6}$$

Then, the feedforward module, consisting of two 1-D convolution (Conv1D) layers with a rectified linear unit (ReLU) activation in between, is applied to further transform the learned features. Specifically, the outputs of the LN operation, i.e., $\mathbf{e}_i^{\text{Trm}}$, are fed to the feedforward module with a residual connection, which is a shared module across all the vectors at each position in a sequence from the MHSA.

Then, LN and residual connection are applied after feedforward to produce the output features of the transformer. Features are fed into a single-layer FC neural network with two linear transformations with an ReLU activation

$$\text{FC}_i = \text{LN}\big(\max\big(W_1\mathbf{e}_i^{\text{Trm}} + b_1, 0\big)W_2 + b_2\big). \tag{7}$$

Finally, the pixel-level prediction for the possibility of the training sample $\mathbf{x}_i$ to be a target (belonging to $\mathbf{X}_T$) or a background sample (belonging to $\mathbf{X}_B$) can be obtained by softmax operation ($\text{FC}_i = [fc_i^1,\ fc_i^2]$)

$$\mathbf{p}_i = \text{softmax}(FC_i) \tag{8}$$

where $p_i^j \in \mathbf{p}_i = [p_i^1,\ p_i^2]$ can be calculated as

$$p_i^j = \frac{e^{f_i^j}}{e^{f_i^1} + e^{f_i^2}}. \tag{9}$$

To calculate the discrepancy between the prediction result and pseudolabel, the loss value for the input pixel sequence $\mathbf{s}_i$ can be computed as

$$\mathcal{L}\big(p_i^1, p_i^2, y_i\big) = -\big[y_i \cdot \log\big(p_i^1\big) + (1 - y_i) \cdot \log\big(p_i^2\big)\big] \tag{10}$$

where $p_i^1$ and $p_i^2$ represent the prediction possibility of sample $\mathbf{x}_i$ to be target and background, respectively.

The $i$th output $y_i$ represents the pseudolabel of sample $\mathbf{x}_i$ as

$$y_i = \begin{cases} 1, & \mathbf{x}_i \in \mathbf{X}_T \\ 0, & \mathbf{x}_i \in \mathbf{X}_B. \end{cases} \tag{11}$$

*2) Distributed Learning Strategy:* To take the unique advantages provided by DDL and solve large-scale problems during training, the distributed learning strategy has been proposed and embedded into HTDFormer.

In distributed learning, the synchronization process plays a crucial role [49], where each worker needs to synchronize gradients with all other workers to update the training model. Recently, the RAR gradient synchronization architecture has become increasingly popular for DDL training [42], which uses ring topology to connect each worker, making sure they only communicate with their neighbors.

In this work, we implement the distributed learning strategy on four workers based on the RAR architecture, as shown in Fig. 2. Specifically, we form a ring between the workers, calculate the gradients of the loss function, and communicate with all other workers to aggregate all the local gradients and update parameters. Each worker executes the training procedure in parallel and periodically synchronizes with other workers. In general, the distributed optimization in HTDFormer can be concluded as the following function:

$$\min \mathcal{L}_{\text{dis}} = \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}_i \tag{12}$$

where $n$ is the number of workers, $\mathcal{L}_i$ is the loss value from worker $i$, and $\mathcal{L}_{\text{dis}}$ is the whole loss. The worker $i$ has a local copy of the model and a partition of the training data, i.e., $\mathbf{S}_i$. And the workers jointly update the model parameters.

Consequently, given a set of input sequence samples in $\mathbf{S}$ with their corresponding pseudolabel defined in (11), the loss value of data partition $\mathbf{S}_i$ computed by worker $i$ can be defined as

$$\mathcal{L}_i = \sum_{j=1}^{M/n} \mathcal{L}\big(p_{i,j}^1, p_{i,j}^2, y_{i,j}\big) = \sum_{j=1}^{M/n} \mathcal{L}\big(f(w_i, \mathbf{s}_{i,j}), y_{i,j}\big) \tag{13}$$

where $\mathbf{p}_{i,j} = [p_{i,j}^1,\ p_{i,j}^2]$ and $y_{i,j}$ represent the pixel-level prediction and the corresponding pseudolabel of the sequence sample $\mathbf{s}_{i,j}$ from data partition $\mathbf{S}_i = [\mathbf{s}_{i,1}; \mathbf{s}_{i,2}; \ldots; \mathbf{s}_{i,M/n}]$, $w_i$ denotes the model parameters of worker $i \in [1, 2, \ldots, n]$, and $f(\cdot)$ indicates the forward inference depicted in Fig. 2(a).

Specifically, in a training epoch, in forward propagation, each worker $i$ performs the forward propagation by calculating the above loss function (10) for each sequence sample in the corresponding data partition $\mathbf{S}_i$, as shown in Fig. 2(a). Subsequently, each worker $i$ performs back propagation to update model parameters by minimizing (13) via a specific optimization algorithm, as shown in Fig. 2(b). Each worker $i$ splits its gradients $g_i$ into $n$ subvectors at first.

The back propagation can be divided into two phases.

1) The first is the scatter–reduce phase, where each worker sends its reduced subvector to its downstream worker, while receiving the reduced subvector from its upstream worker to compute a new reduced subvector.
2) The second phase is the all-gather phase, where each worker sends its newly received fully reduced subvector to its downstream worker, while receiving a new fully reduced subvector from its upstream worker.

Therefore, in the proposed framework with $n$ workers, $2(n-1)$ communication steps are required by RAR for completing the synchronization process. Since each worker sends $(g_i/n)$ amount of gradients for $2(n-1)$ times, the total amount of gradients each worker sends is $(2g_i(n-1)/n)$, which is asymptotically independent of $n$ as $n$ gets large. In this work, the number $n$ has been set as four.

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**Distributed Learning**:
For $n$ workers, default settings are $m_0 = 0$, $v_0 = 0$, $\eta = 0.003 \cdot n$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\xi = 10^{-6}$. $\tilde{g}_t^2$ denotes the elementwise square $\tilde{g}_t \odot \tilde{g}_t$, $g_0 = 0$. Each worker has the same view of the model.

**Input**:
$m_0$: Initial $1^{st}$ moment vector
$v_0$: Initial $2^{nd}$ moment vector
$\eta$ : Learning rate
$\beta$: Exponential decay rates for moment estimates
$\xi$: Weight coefficient
$r$: Compression ratio
$thr$: Selection threshold
$\tilde{g}_t$: Compressed gradients

**for** $t = 1$**to**$T$ **do**
**from** worker $i = 1$ **to** worker $i = n$ **do in parallel**:
Worker $i$ draw sequences samples $\mathbf{s}_{i,j}$ from $\mathbf{S}_i$.
Worker $i$ computes ($\mathcal{L}_i$ is defined in (13)):
$g_t \leftarrow g_{t-1}$
$g_t = g_t + \frac{1}{|\mathbf{S}_i|} \sum_{\mathbf{s}_{i,j} \in \mathbf{S}_i} \nabla \mathcal{L}_i \left(w_{i,t}, \mathbf{s}_{i,j}\right)$
Select threshold: $thr \leftarrow r\%$ of $|g_t|$
$b \leftarrow |g_t| > thr$
$\tilde{g}_t \leftarrow g_t \odot b$
$g_t \leftarrow g_t \odot \neg b$
Worker $i$ splits gradients $\tilde{g}_t$ into $n$ subvectors and updates $\tilde{g}_t$ under RAR rule (see Fig. 2(b)).
Worker $i$ updates parameters:
$m_t = \beta_1 m_{t-1} + (1 - \beta_1)\tilde{g}_t$
$v_t = \beta_2 v_{t-1} + (1 - \beta_2)\tilde{g}_t^2$
$m_t = m_t / \left(1 - \beta_1^t\right)$
$v_t = v_t / \left(1 - \beta_2^t\right)$
$u_t = \frac{m_t}{\sqrt{v_t} + \xi}$
$w_{i,t+1} = w_{i,t} - \phi\left(\left\|w_{i,t}\right\|, \|u_t\|\right) u_t \eta$
**end for**

On this basis, with the introduction of data parallelism, the batch size should be large enough to use the computational resources of the workers. Therefore, the large-batch training strategy has been designed to exploit large-scale distributed processing via the above-distributed learning strategy. Specifically, we adopt the layerwise adaptive large batch optimization technique (LAMB) [50], as shown in Distributed Learning, based on the ADAM [51] algorithm which is popular in DL and has shown to have good performance for training language models [50]. Specifically, the weights' update rule of a worker for all the layers of the network during training can be concluded as

$$w_{t+1} = w_t - \phi(\|w_t\|, \|u_t\|) u_t \eta \tag{14}$$

where $w_t \in \mathbb{R}^d$ and $u_t$ are the model parameters and the update of corresponding gradients at time step $t$, $\eta$ is the learning rate, $\|\cdot\|$ represents the $l2$-norm operation, and $\phi(a, b)$ is the scaling function which can be demonstrated as

$$\phi(a, b) = \begin{cases} 1, & a \le 0 \\ \dfrac{a}{b}, & a > 0,\ b > 0 \\ 1, & a > 0,\ b \le 0. \end{cases} \tag{15}$$

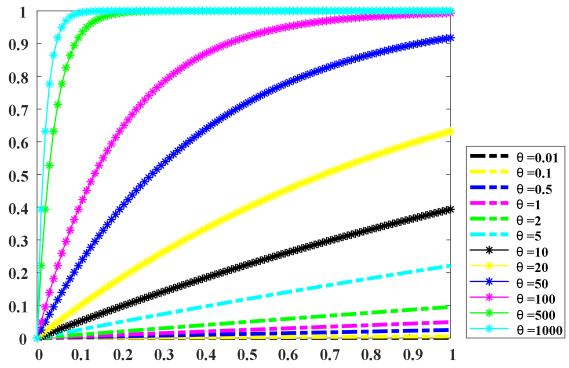



Fig. 3. Shape of exponential constrained nonlinear transform factor $\kappa$.

Considering that the above-mentioned communication between distributed workers involves large-batch training, we use the gradient compression strategy to reduce communication cost for the proposed distributed learning framework.

In this work, we use the deep gradient compression (DGC) algorithm [52], a representative sparsification method, to compress the gradients for efficient exchange among workers. Let $g_{i,t}$ be the local gradients in worker $i$ during training iteration $t$. The DGC compressor selects only a subset of the elements of the original gradients $g_{i,t}$ in an epoch, resulting in a sparse vector for correspondence. Meanwhile, the other gradients are preserved for accumulation via a memory mechanism to compensate for the update of the gradients at the next epoch.

Specifically, in an epoch, each worker locally computes a gradient $g_{i,t}$ based on a batch of training samples at first. Then, worker $i$ combines $g_{i,t}$ with its memory $m_{i,t-1}$ from the previous iteration to update the local gradients $g_{i,t}$ at this iteration. Next, the worker produces compressed gradients $\tilde{g}_{i,t}$ and updates the memory $m_{i,t}$ by applying compression under a selection rule with a user-defined compression ratio $r$. According to $r$, for all the elements in $g_{i,t}$, worker $i$ adaptively determines the corresponding threshold thr and a mask vector $b$ with the same number of elements as $g_{i,t}$. In detail, the value 1 in $b[k]$ indicates that the corresponding gradient element $g_{i,t}[k]$ is selected. The elementwise product $g_{i,t} \odot b$ generates a sparse vector $\tilde{g}_{i,t}$ of the original gradient $g_{i,t}$. Consequently, only gradient elements $g_{i,t}[k] < -\text{thr}$ and $g_{i,t}[k] > \text{thr}$ in $\tilde{g}_{i,t}$ are transmitted under RAR rule to finish the weights' update.

### C. Testing via Spectral–Spatial Fusion

After distributed training, each testing pixel is processed by the spectral sequence construction procedure to produce a testing sequence. Then, the sequences are further processed by the SE, PE, and trained transformer encoder to realize pixel-level inference and produce pixel-level prediction $\mathbf{R}$. Although the prediction $\mathbf{R}$ based on transformer appears with high recall rate, the false alarm needs to be reduced.

Considering the extracted spectral and spatial information are complementary to each other, spectral–spatial fusion is proposed. Specifically, the final detection result $\mathbf{Detc} \in \mathbf{R}^{M \times N}$ can be produced based on $\mathbf{R}$ by an exponential constrained

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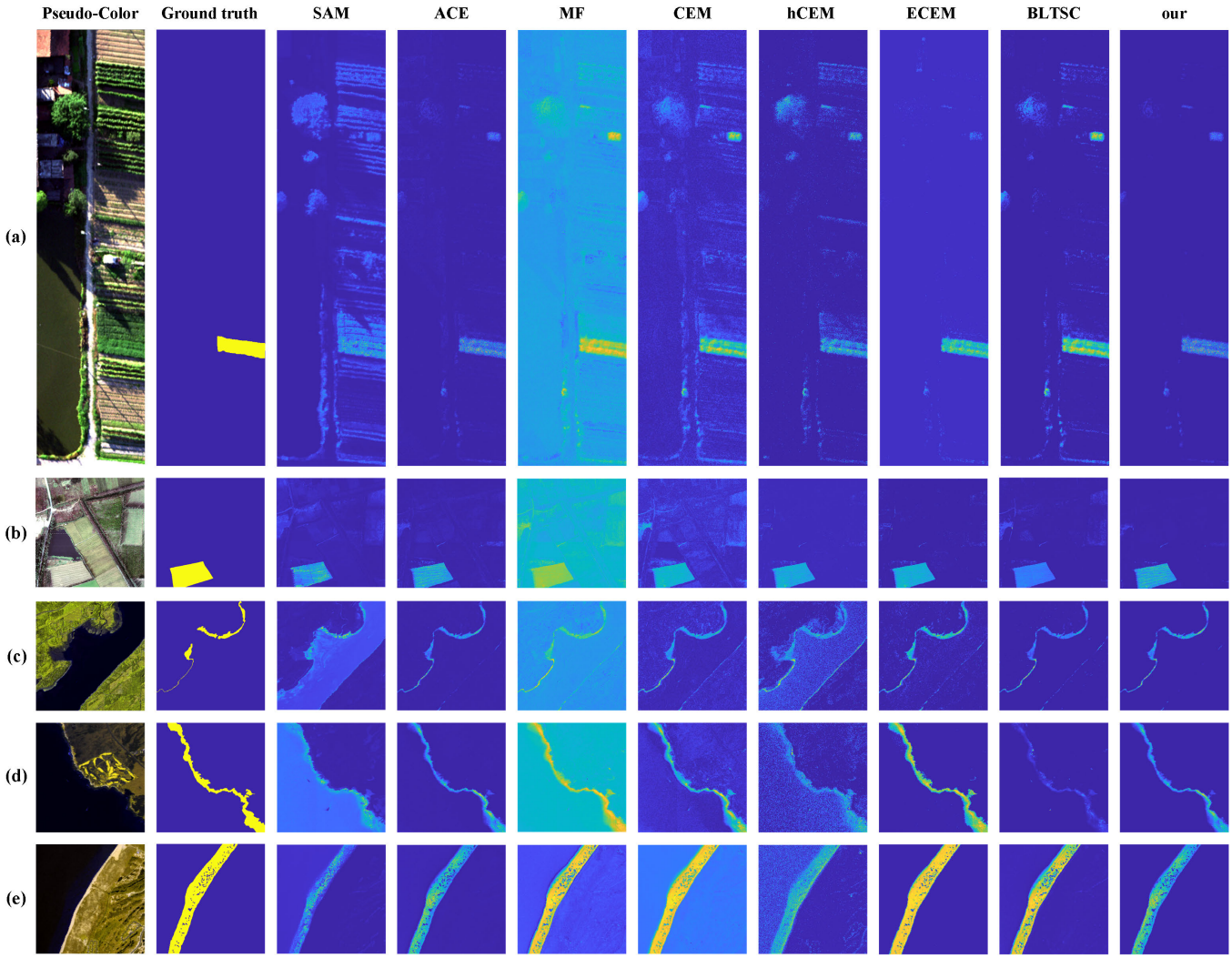



Fig. 4. Detection results of eight algorithms on five large-scale HSIs: (a) HanChuan, (b) Matiwan, (c) HNU-1, (d) HNU-2, and (e) HNU-3.

nonlinear function as follows:

$$\mathbf{Detc} = \boldsymbol{\kappa} \odot \mathbf{R} = \left(\mathbf{1} - e^{-0.05 \cdot \theta \cdot \mathbf{C}}\right) \odot \mathbf{R} \tag{16}$$

where $\theta$ is a positive parameter to adjust the shape of nonlinear transformation factor $\boldsymbol{\kappa}$, which is proposed to use the spatial map $\mathbf{C}$ from coarse detection to suppress the background and enhance the detection performance.

The instances with high likelihood of being target according to spectral–spatial similarity are preserved, and $\boldsymbol{\kappa}$ is relatively stable and approximates to 1. On the contrary, those with a high likelihood of being background are suppressed, and $\boldsymbol{\kappa}$ falls sharply and approximates to 0. Fig. 3 shows the shape of $\boldsymbol{\kappa}$ under different choices of $\theta$. Thus, the spectral–spatial fusion procedure suppresses the background.

## III. Experimental Analysis

This section introduces the experimental environment for performance evaluation, the information about the hyperspectral datasets, and parameter settings for the compared algorithms, followed by thorough evaluation results to justify the proposed approach in terms of detection performance computational efficiency. The proposed HTD framework was implemented and evaluated on a distributed system built on a server equipped with an 18-core Intel Core i9-10980XE CPU operating at 3.00 GHz and 128-GB RAM. The distributed system has installed Ubuntu 18.04 as the operating system and consists of four computing workers. Each worker is equipped with an NVIDIA Geforce RTX 3090 GPU with 24-GB memory. All the testing experiments are conducted with a single worker.

### A. Experimental Setup

*1) Datasets:* The proposed HTDFormer is evaluated on three real HSI datasets including HanChuan, Matiwan, and HNU. These datasets are acquired at various scenes by different sensors, in which multiscale targets of different categories are distributed. Fig. 4 displays the pesudocolor representation of these datasets with the corresponding ground-truth maps.

The HanChuan dataset, an HSI of the WHU-Hi dataset [53], was acquired in Hanchuan, Hubei province, China, with a 17-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a Leica Aibot X6 UAV V1 platform.

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The size of the HSI is 1217 × 303 pixels. There are 274 bands from 400 to 1000 nm. And the spatial resolution of the UAV-borne hyperspectral imagery is about 0.109 m. The sorghum in this dataset is selected as the target category. Meanwhile, the background contains many other crop species: strawberry, cowpea, soybean, water spinach, watermelon, and greens.

The Matiwan dataset, cropped from the Xiongan New Area dataset [54], was captured with a wavelength from 400 to 1000 nm by the full-spectrum multimodal imaging spectrometer developed by the Shanghai Institute of Technical Physics, Chinese Academy of Sciences. The HSI has 256 available bands with a spatial resolution of 0.5 m. The spatial pixel size of the HSI is 790 × 750. The vegetable field in this dataset is selected as the target category. The background contains sophora japonica, paddy, soybean, and water field.

The HNU dataset consists of three HSIs from the HNU-UTD dataset [55], which are denoted as HNU-1, HNU-2, and HNU-3. These HSIs were captured by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS). After removing the invalid bands mainly resulted by water absorbing, 177 bands were kept for these three HSIs. The spatial pixel size of HNU-1, HNU-2, and HNU-3 is 300 × 300, 230 × 230, and 300 × 300, respectively. The spatial resolution HNU-1, HNU-2, and HNU-3 is 16.4, 17, and 3.4 m, respectively. The target category of HNU-1, HNU-2, and HNU-3 is sandstone, rock, and silt sediment, respectively.

*2) Methods:* Seven popular HTD methods mentioned above were adopted for comparison, including SAM [4], ACE [7], MF [9], CEM [8], hCEM [10], E-CEM [11], and BLTSC [56], which are either frequently cited in the literature or offer leading performances. Among these methods, the SAM [4], ACE [7], MF [9], and CEM [8] are the typical traditional HTD algorithms. The hCEM [10] and the E-CEM [11] are the popular extended versions of the CEM, which have been widely used. The BLTSC [56] is a representative HTD model based on DL, which has adopted coarse detection based on CEM.

In addition, here are the parameter settings of all the methods for testing except for the SAM, ACE, MF, and CEM without adjustable parameters. For the hCEM, the parameters $\lambda$ and $\epsilon$ are set as 100 and $1e$-4, respectively. For the E-CEM, the number of detection layers, the number of CEM detectors per layer, and the regularization coefficient are set as 10, 6, and $1e$-6, respectively. For the BLTSC, the parameters are consistent with those set in [56], which trains model for 500 epochs with the learning rate set as $1e$-4 for all the datasets. For the proposed HTDFormer, the selection ratio $\sigma\%$ is set as 1%, 5%, 2%, 5%, and 1% for HanChuan, Matiwan, HNU-1, HNU-2, and HNU-3, respectively. The adjustable parameter $\theta$ in nonlinear transformation is set as 100, 20, 20, 1, and 20 for HanChuan, Matiwan, HNU-1, HNU-2, and HNU-3, respectively. Besides, the number of heads in MHSA, i.e., $m$, is set as 1 for all the datasets. The number of training epochs is set as 1600, 500, 1500, 2000, and 2000 for HanChuan, Matiwan, HNU-1, HNU-2, and HNU-3, respectively.

*3) Metrics:* The receiver operating characteristic (ROC) [57] curve and the area under the ROC curve (AUC) [58] are adopted to qualitatively and quantitatively evaluate the performance of models. The ROC is a widely used evaluation metric in target detection, which describes the varying relationship between the true-positive rate (TPR) and false-positive rate (FPR) based on the reference map, and can be estimated via plotting the TPR and FPR at various thresholds denoted as $\tau$. The AUC metric is regarded as a complete description of detection performance, which is derived from the ROC curve by calculating the whole area under the ROC curve. Specifically, $\mathrm{AUC}_{(D,F)}$ and $\mathrm{AUC}_{(F,\tau)}$ are indicators of how effective a detector is and how high the detection probability and how well the background suppressibility of a detector can achieve. The higher the value of $\mathrm{AUC}_{(D,F)}$ and the smaller the value of $\mathrm{AUC}_{(F,\tau)}$, the better the detector. Note that we use $\mathrm{AUC}_1$ and $\mathrm{AUC}_2$ to refer to the AUC value of (FPR, TPR) and (FPR, $\tau$), respectively. On the basis, the target–background separability map [59], the convergence result [60], and the communication-learning tradeoff [61] are adopted to further evaluate the proposed framework.

### B. Detection Performance

Fig. 4 visualizes the detection results of eight algorithms. It is obvious that SAM, MF, and CEM perform much worse than others in background suppression, especially on the HanChuan and Matiwan datasets. HTDFormer obtains much more convincing target detection results than hCEM, as well as better background suppression performance than E-CEM and BLTSC. Although ACE may obtain relatively good performance in target recognition, it cannot suppress background well in every scene. We can observe that the distinction between the target and background of HTDFormer is much larger than that of ACE, especially on the HanChuan dataset. Even though BLTSC based on background distribution estimation provides the best results in background suppression, it omits too many interesting target pixels, especially in Matiwan and HNU-2. In comparison to other methods, HTDFormer generates the detection results much more similar to the ground truth. Therefore, the proposed framework shows a better capability to suppress background than most competitors under high detection reliance.

Consistent with Fig. 4, Table I provides AUC indicators of all the comparing methods on five large-scale HSIs for quantitative evaluation. A detector with a larger AUC1 value is deemed to have a better detection performance. HTDFormer obtains the highest AUC1 value on Matiwan and HNU-1, as well as competitive AUC1 values on other HSIs, which shows the effectiveness of HTDFormer in target detection. Meanwhile, when AUC1 scores of several methods are similar, AUC2 scores are used to evaluate these methods. In AUC2 value, HTDFormer always offers the best competitiveness, which verifies its overall effectiveness on HTD tasks. A smaller AUC2 value indicates that the background is suppressed better in the detection results [3]. Therefore, the proposed HTDFormer shows a better capability to suppress background than most competitors under high detection reliance.

Fig. 5(a)–(e) shows ROC curves of (FPR, TPR) corresponding to the above detection results. Note that the closer to

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TABLE I
AUC SCORES OF THE ALGORITHMS ON FIVE LARGE-SCALE HSIS

| Methods | HanChuan | | Matiwan | | HNU-1 | | HNU-2 | | HNU-3 | |
|---|---|---|---|---|---|---|---|---|---|---|
| | $AUC_1$ | $AUC_2$ | $AUC_1$ | $AUC_2$ | $AUC_1$ | $AUC_2$ | $AUC_1$ | $AUC_2$ | $AUC_1$ | $AUC_2$ |
| SAM | 0.98327 | 0.03287 | 0.98696 | 0.02126 | 0.97013 | 0.06407 | 0.91685 | 0.11950 | 0.98228 | 0.01940 |
| ACE | 0.99193 | 0.00585 | 0.99376 | 0.01046 | 0.99316 | 0.00571 | 0.99291 | 0.00510 | 0.99539 | 0.01000 |
| MF | **0.99520** | 0.36847 | 0.99217 | 0.44906 | 0.99309 | 0.33601 | 0.99700 | 0.45945 | 0.99946 | 0.24869 |
| CEM | 0.99416 | 0.05021 | 0.99146 | 0.06729 | 0.99199 | 0.03999 | 0.99349 | 0.05310 | 0.99678 | 0.12270 |
| hCEM | 0.94639 | 0.01177 | 0.99246 | 0.02879 | 0.95498 | 0.06749 | 0.92083 | 0.05810 | 0.97473 | 0.06402 |
| ECEM | 0.98729 | 0.04089 | 0.99318 | 0.00371 | 0.99219 | 0.01605 | **0.99713** | 0.01557 | **0.99956** | 0.01090 |
| BLTSC | 0.99391 | 0.00961 | 0.99161 | 0.00676 | 0.99079 | 0.00439 | 0.98777 | 0.00182 | 0.99734 | 0.01077 |
| HTDFormer | 0.99430 | **0.00393** | **0.99483** | **0.00336** | **0.99488** | **0.00340** | 0.99389 | **0.00178** | 0.99700 | **0.00177** |

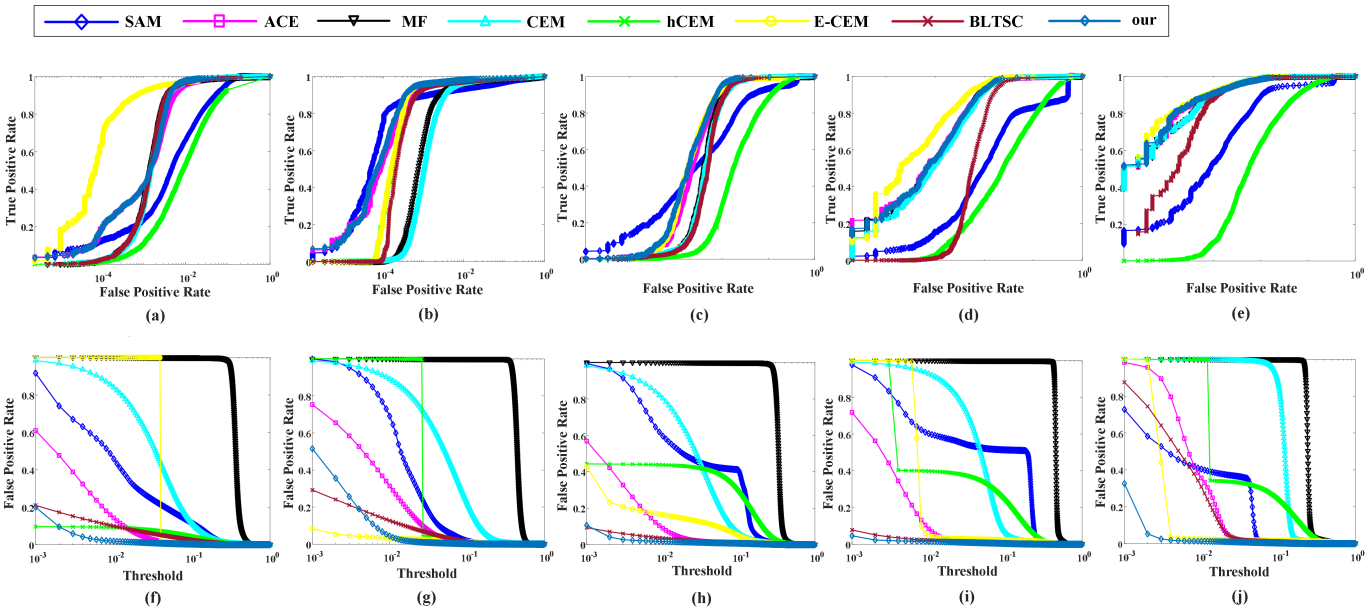



Fig. 5. First row shows the ROC curves of (FPR,TPR) for eight algorithms on five HSIs: (a) HanChuan, (b) Matiwan, (c) HNU-1, (d) HNU-2, and (e) HNU-3. The second row shows the ROC curves of (FPR, $\tau$) for eight algorithms on five HSIs: (f) HanChuan, (g) Matiwan, (h) HNU-1, (i) HNU-2, and (j) HNU-3.

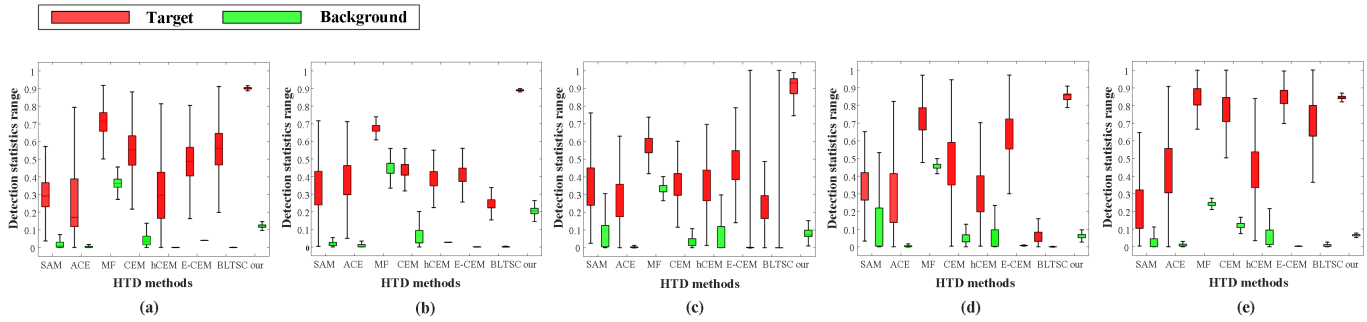



Fig. 6. Separability maps for eight algorithms on five large-scale HSIs: (a) HanChuan, (b) Matiwan, (c) HNU-1, (d) HNU-2, and (e) HNU-3.

the upper left corner of a curve, the better the performance. It can be observed that the ROC curve of HTDFormer is closer to the upper left corner than that of other methods on Matiwan and HNU-1, but performs slightly worse than on other HSIs. Meanwhile, as shown in Fig. 5(f)–(j), the ROC curves of (FPR, $\tau$) of the proposed method are much closer to the lower left corner than other methods on all the datasets, which shows better performance of the proposed method.

Furthermore, the target–background separability map is adopted to compare the detection performance, as shown in Fig. 6. The larger distance between the lower bound of the red box and the upper bound of the green box indicates better target–background separability of HTDFormer.

Besides, the testing time of SAM, ACE, MF, CEM, hCEM, E-CEM, BLTSC, and HTDFormer is summarized in Table II. Although HTDFormer is not the most efficient method, it is much faster than most comparing methods except for only

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TABLE II
COMPUTING TIME (IN SECONDS) OF METHODS IN COMPARISON

| Methods | HanChuan | Matiwan | HNU-1 | HNU-2 | HNU-3 | Average |
|---|---|---|---|---|---|---|
| SAM | **2.5604** | 4.2267 | 0.6050 | 0.3499 | 0.6054 | 1.6695 |
| ACE | 8.8442 | 8.2531 | 0.3418 | 0.2154 | 0.3439 | 3.5997 |
| MF | 2.6092 | **3.4388** | **0.1544** | **0.0939** | **0.1553** | **1.2903** |
| CEM | 5.0344 | 11.4369 | 4.2774 | 2.6667 | 4.1511 | 5.5133 |
| hCEM | 19.2158 | 23.6693 | 3.6962 | 2.2377 | 3.3890 | 10.4416 |
| ECEM | 477.7106 | 631.2875 | 33.8025 | 25.345 | 33.8656 | 152.6283 |
| BLTSC | 8.9643 | 16.6983 | 6.3659 | 3.2062 | 5.2258 | 8.0921 |
| HTDFormer | 4.3738 | 9.3678 | 1.1776 | 0.6570 | 0.9934 | 3.3139 |

two traditional models, i.e., SAM and MF. Compared with the DL methods such as BLTSC, HTDFormer not only realizes improvement in detection performance but also remarkably reduces computational complexity. The first reason why HTDFormer is relatively fast in prediction is due to the relatively shallow architecture, and unlike other DL methods, it does not require a very deep architecture to achieve satisfactory performance. Moreover, the second reason is due to the parallel implementation of each head in the MHSA mechanism of transformer especially since only one head has been adopted on the testing datasets.

From the above experimental results, we can conclude that the introduction of transformer helps HTD model effectively extract spatial similarity for target–background separation without excessive computing consumption. Furthermore, compared with other conventional methods and learning-based approaches, the proposed framework using pseudosamples, consistent with the characteristics of WSL, has performed well on the inferior samples.

### C. Ablation Experiment

In this section, we have assessed the effectiveness of different components of the proposed framework. Note that the sample augmentation procedure can be regarded as a fixed part of the use of transformer because transformer cannot be directly applied in HSIs.

Table III shows the influence on detection performance from the introduction of the transformer and the spectral–spatial inference fusion. Specifically, the experiments are conducted on different datasets based on different coarse detectors. In terms of coarse detectors, we evaluated the proposed transformer-based model based on different coarse detectors. Considering that the coarse detector needs to be as simple and efficient as possible, we choose the ACE algorithm and the SAM algorithm, respectively, to conduct ablation experiments on the datasets of different scales.

As for the dataset, we divided the HSIs into three scales according to the number of pixels, that is, the number of test samples. The large-scale dataset consists of the HanChuan dataset with $1217 \times 303$ pixels and the Matiwan dataset with $790 \times 750$ pixels. The medium-scale dataset consists of the HNU-1 dataset with $300 \times 300$ pixels, the HNU-2 dataset with $230 \times 230$ pixels, and the HNU-3 dataset with $300 \times 300$ pixels. In particular, two additional small HSIs are selected to form the small-scale dataset. The first one is the Bay Champagne dataset with $100 \times 100$ pixels, 188 bands, and 11 target pixels of interest, which was captured by the AVIRIS sensor. The other one is the Pavia dataset with $150 \times 150$ pixels, 102 bands, and 68 target pixels, which was captured by the Reflective Optics System Imaging Spectrometer (ROSIS).

As shown in Table III, the proposed framework with transformer outperforms the ACE algorithm and algorithm absolutely in target detection, which is illustrated in $AUC_1$ value. Meanwhile, the spectral–spatial fusion significantly strengthens the background suppression of the whole framework, which is illustrated in $AUC_2$ value. The ablation study points out that the original coarse detectors can detect potential targets from spectral information well. However, they cannot make full use of the spatial correlation among pixels in HSIs. On the basis of coarse detection, the transformer encoder in the proposed framework realizes better target–background separation via extracting spectral–spatial features. Therefore, it can be concluded that the transformer encoder makes full use of global spatial information and produces detection results with complementary information to coarse detection. Finally, the spectral–spatial fusion imposes joint spectral–spatial inference result from transformer on the coarse detection result. This procedure effectively suppresses the background by combining the discriminative feature from the spectral and spatial domains.

Table IV provides the training times using different numbers of workers on the proposed HTDFormer framework. The time statistics in columns three to six of Table IV show that the total training time decreases as the number of workers grows. Specifically, referring to the last row statistics reported in Table IV, we observe that the speedup shows roughly a linear increase with regard to worker count. Therefore, the embedding of distributed learning strategy can dramatically reduce the amount of time needed to perform the full training process, which opens up more flexible ways to study HTD, especially in processing large-scale HSIs.

Fig. 7 depicts the training loss curves of training HTDFormer with different optimizers at varying training epochs. Specifically, the Adagrad [62], Adam [51], and AdamW [63] are used to compare with the adopted LAMB optimizer. The Adagrad [62] adapts the learning rate to the parameters by performing larger updates for infrequent and smaller updates for frequent parameters. The Adam [51] stores both an exponentially decaying average of past squared gradients and an exponentially decaying average of past gradients. The AdamW [63] improves regularization in the Adam by decoupling the weight decay from the gradient-based update [63]. Taking the curve on HanChuan in Fig. 7(a) as an example, the LAMB optimizer works better obviously than other optimizing algorithms, especially for large-scale real HSIs. When the training epoch is 100, it is observed that LAMB achieves roughly the same training loss as others. However, as the training epoch grows to 300, LAMB achieves visibly smaller training loss than others. This observation illustrates that LAMB improves large-scale distributed processing and boosts convergence during training.

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TABLE III
INFLUENCE FROM THE PROPOSED COMPONENTS ON HSIS WITH DIFFERENT SCALES IN THE AUC SCORES

| Scale of Dataset | Coarse Detector | Components | HanChuan | | Matiwan | |
|---|---|---|---|---|---|---|
| | | | $AUC_1$ | $AUC_2$ | $AUC_1$ | $AUC_2$ |
| Large | ACE | Coarse Detection | 0.99193 | 0.00585 | 0.99376 | 0.01046 |
| | | our − fusion | **0.99592** | 0.13757 | **0.99805** | 0.21630 |
| | | our + fusion | 0.99430 | **0.00393** | 0.99483 | **0.00336** |
| | SAM | Coarse Detection | 0.98327 | 0.03287 | 0.98696 | 0.02126 |
| | | our − fusion | 0.99235 | 0.08548 | 0.99022 | 0.13517 |
| | | our + fusion | **0.99700** | **0.00627** | **0.99239** | **0.00442** |

| Scale of Dataset | Coarse Detector | Components | HNU-1 | | HNU-2 | | HNU-3 | |
|---|---|---|---|---|---|---|---|---|
| | | | $AUC_1$ | $AUC_2$ | $AUC_1$ | $AUC_2$ | $AUC_1$ | $AUC_2$ |
| Medium | ACE | Coarse Detection | 0.99316 | 0.00571 | 0.99291 | 0.00510 | 0.99539 | 0.01000 |
| | | our − fusion | **0.99743** | 0.09961 | 0.99168 | 0.07179 | **0.99746** | 0.07572 |
| | | our + fusion | 0.99488 | **0.00340** | **0.99389** | **0.00178** | 0.99700 | **0.00177** |
| | SAM | Coarse Detection | 0.97013 | 0.06407 | 0.91685 | 0.11950 | 0.98228 | 0.01940 |
| | | our − fusion | 0.99304 | 0.13540 | 0.98484 | 0.07535 | 0.99186 | 0.01895 |
| | | our + fusion | **0.99548** | **0.01667** | **0.98847** | **0.01178** | **0.99275** | **0.00122** |

| Scale of Dataset | Coarse Detector | Components | Bay Champagne | | Pavia | |
|---|---|---|---|---|---|---|
| | | | $AUC_1$ | $AUC_2$ | $AUC_1$ | $AUC_2$ |
| Small | ACE | Coarse Detection | 0.92239 | 0.00746 | 0.77501 | 0.03466 |
| | | our − fusion | **0.99973** | 0.32049 | **0.96456** | 0.26128 |
| | | our + fusion | 0.97086 | **0.00357** | 0.87184 | **0.01255** |
| | SAM | Coarse Detection | 0.95859 | 0.08045 | 0.82472 | 0.24137 |
| | | our − fusion | 0.99425 | 0.06541 | **0.94567** | 0.35162 |
| | | our + fusion | **0.99691** | **0.01210** | 0.89871 | **0.15789** |

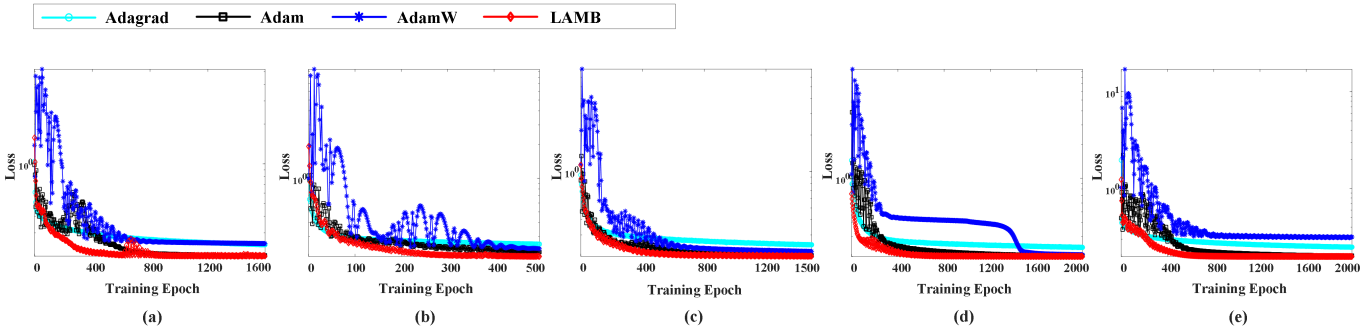



Fig. 7. Convergence results from different optimizers including Adgard, Adam, AdamW, and LAMB with respect to training loss on five HSIs: (a) HanChuan, (b) Matiwan, (c) HNU-1, (d) HNU-2, and (e) HNU-3.

TABLE IV
TRAINING TIME STATISTICS (IN SECONDS) OF HTDFORMER FOR VARIOUS DATA SIZES WITH DIFFERENT NUMBERS OF WORKERS

| Training data | Data size | Training time (s) | | | |
|---|---|---|---|---|---|
| | | 1 worker | 2 workers | 3 workers | 4 workers |
| HanChuan | $(1217, 303, 274)$ | 944.98 | 522.25 | 364.86 | 288.74 |
| Matiwan | $(790, 750, 256)$ | 1087.27 | 623.02 | 429.10 | 297.41 |
| HNU-1 | $(300, 300, 177)$ | 258.66 | 156.79 | 118.62 | 83.81 |
| HNU-2 | $(230, 230, 177)$ | 297.62 | 186.33 | 126.72 | 98.26 |
| HNU-3 | $(300, 300, 177)$ | 717.14 | 400.01 | 271.30 | 209.39 |
| Total | | 3305.67 | 1888.4 | 1310.6 | 977.61 |
| Speedup | | 1 × | 1.75 × | 2.52 × | 3.38 × |

Fig. 8 shows the tradeoff between communication cost and learning performance in terms of detection accuracy on different HSIs. We compare this tradeoff among the adopted DGC algorithm, the Top-$k$ [64] algorithm, and the Random-$k$ [65] algorithm. The Top-$k$ [64] sparsifies gradient updates by removing a specific ratio of smallest gradients by absolute value, dubbing this gradient dropping. Based on random sparsification, Random-$k$ [65] shows an advantage over top-$k$ sparsification especially in the multicore shared memory implementation. Besides, we also list the accuracy achieved by the same optimizer without gradient compression as a benchmark. In Fig. 8, all the four situations show a communication-learning tradeoff, that is, the more communication cost can be used, the better detection performance can be achieved. Furthermore, it can be concluded that when the communication cost is small, increasing the communication cost can bring significant improvement. Compared with no gradient compression, the use of all three compressors can achieve competitive detection performance while effectively reducing the communication cost. On this basis, DGC can

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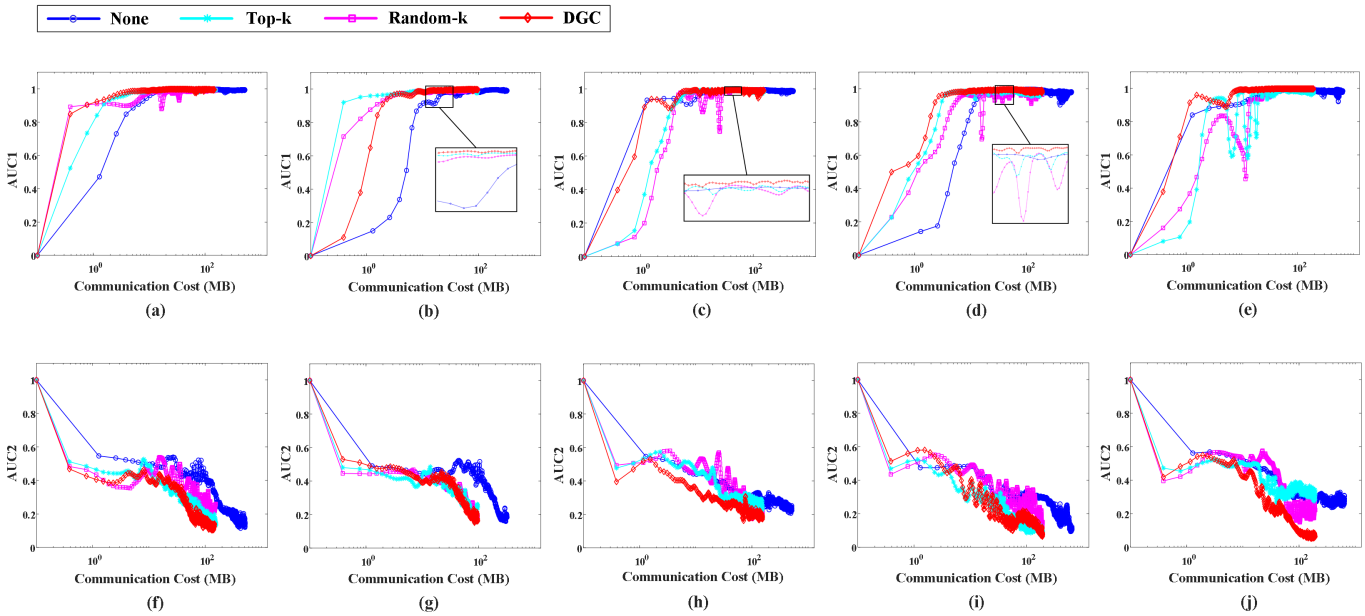



Fig. 8. Communication–learning tradeoffs between detection performance and communication cost from different compressors including Top-$k$, Random-$k$, DGC, and no compressor. The first row shows the change in AUC1 metric on five HSIs: (a) HanChuan, (b) Matiwan, (c) HNU-1, (d) HNU-2, and (e) HNU-3. The second row shows the change in AUC2 metric on five HSIs: (f) HanChuan, (g) Matiwan, (h) HNU-1, (i) HNU-2, and (j) HNU-3.

achieve a higher $AUC_1$ value and better $AUC_2$ value than the other two methods under the same communication cost.

## IV. Conclusion

This article presents a new weakly supervised HTD framework named HTDFormer based on transformer with distributed learning for HTD tasks in a distributed fashion. Specifically, HTDFormer realizes the spectral–spatial joint optimization via transformer to solve the weakness of the existing HTD methods only using spectral characteristics, especially in large-scale hyperspectral scenes. The proposed sample augmentation procedure can use the spectrums of each pixel and its neighboring pixels to produce sequences as the input of transformer. For the first time, using the RAR distributed architecture, we demonstrate a transformer-based HTD framework with DDL embedded in it. Finally, the testing procedure combines the feature from the coarse detection and transformer inference via an exponential constrained nonlinear function, which further addresses the issues of insufficient utilization of spatial information. The experiments carried out in this article demonstrate that HTDFormer provides efficient and effective detection performance. In addition, the obtained results reveal that the distributed solution can enable us to successfully exploit large-scale distributed processing in HTD tasks. In the future, we will develop other implementations of HTD techniques in distributed environments to offer the possibility to process massive amounts of pseudosamples in a distributed way.

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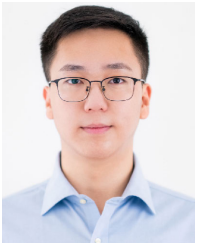
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